Structural certificates for the $\mathrm{Sp}(6,2)$ atlas table

L054 finite application

Theorem 1. *Relative to the acquired numerical catalogue and named structural registry, all* 1369 *rows of the current companion table headed* $\mathrm{Sp}(6,2)$ *have the displayed structural descriptions. The ambient group is* $\mathrm{PSp}_6(2)$*, of order* 1451520*, in the acquired action on* 63 *points. An extension description asserts the indicated normal subgroup, quotient and splitting behaviour. It does not determine an abstract isomorphism class uniquely from the printed expression.*

Source and acquired results. The source is the Connor–Leemans companion table dated 7 June 2013, associated with [1]. Its numerical catalogue, source-row bijection, actual embedded subgroup generators and named top group are acquired from the separately accepted numerical computation. No new ambient subgroup census is used here. The 515 acquired structural models come from earlier accepted finite applications. Each of the 52 files read to reconstruct this registry agrees with its entry in the accepted $\mathrm{Sp}(4,7)$ review's dependency record. Acquiring these inputs retains their existing hypotheses and qualifications.

The paper's Section 3.2 adopts left association for an unparenthesized sequence of extensions. We have read that convention on native printed page 263. A position audit joins the powers, subscripts, operators, parentheses and wrapped structural cells of all 1369 rows to their parsed expressions and orders. Direct inspection of the companion covers pages $1, 2, 49, 86, 87$. It is not an inspection of every page. The position audit uses the same pypdf library as the flat extraction, so these are not two independent extraction engines. The exact retained files, rather than an assertion of identity with every historical edition, specify the source.

What is supplied by this application. There are 199 new finite model certificates: 143 split extensions, 35 direct products and 21 nonsplit extensions. Every proper source occurrence has a transport from a proved model to its acquired subgroup of the 63-point action. The source row, its literal expression, its order and its actual generators are joined before the transport is checked. Candidate models used to help later searches are still proved individually; they are never promoted to acquired facts merely because a search returned them. All references to new factor models precede the model using them.

Lemma 1. *Let* M *and* H *be finite groups given by complete closures of specified permutation generators. A bijection* $f : M \to H$ *satisfying* $f(1) = 1$ *and* $f(xg) = f(x)f(g)$ *for every* $x \in M$ *and every specified generator* g *is an isomorphism.*

Proof. Every element $y \in M$ is a positive word in the generators: in a finite group, an inverse of a generator is a nonnegative power of that generator. Induction on such a word gives $f(xy) = f(x)f(y)$ for every $x, y \in M$. Thus f is a homomorphism, and the assumed bijection makes it an isomorphism. The reader constructs the proposed images along the complete closure, checks their consistency whenever a word reaches an old element, and then checks every right-generator edge. Bijectivity is checked against the whole target subgroup, not merely against its generating set. □

Extension certificates. For each new model H, the reader constructs the subgroup N generated by the displayed permutations, verifies $N \triangleleft H$, and transports an earlier model onto N. For

a split extension it similarly verifies a subgroup K, its structure, $N \cap K = 1$ and $|N||K| = |H|$. Normality makes NK a subgroup, and the order equation gives $H = NK$. This proves the semidirect product assertion. For a colon the retained convention also requires nontrivial actual conjugation of K on N; a pair of noncommuting generators supplies the witness. For a direct product all generator pairs commute, hence all elements of the two factors commute. These are the previously accepted finite extension predicates.

The nonsplitting certificates. Thirteen models have prime quotient. If that quotient has order p, a complement would be generated by an element of order p outside N. Conversely such an element generates a complement by the order formula. The reader checks every outside element and excludes this possibility. One model has an abelian quotient. Its certificate gives complete, disjoint normal forms for the quotient and checks every choice of lifts of its stated cyclic generators. Every choice fails an order relation or a commutation relation, so none defines a section.

Five models have quotient S_3. The certificate identifies quotient generators satisfying $a^2 = b^3 = (ab)^2 = 1$ and gives all their lifts. A complement would provide lifts satisfying these same relations. First discarding lifts which fail $a^2 = 1$ or $b^3 = 1$ preserves every possible section; the remaining pairs are checked against $(ab)^2 = 1$. The source count records the full original Cartesian product, including discarded pairs. This filtering is inherited from the earlier accepted reader.

The remaining two dots use the previously accepted Cayley-graph criterion. The quotient is identified with an acquired finite model. For each choice of lifts of its ordered generators, the reader propagates a proposed section along the complete quotient Cayley graph. Consistency on every edge would give a homomorphism by the lemma, and the quotient images would make it a section. Conversely a section supplies such a consistent choice. One inconsistent edge for every lift choice therefore proves that no complement exists. The present instances require 128 lift choices and 2176 edge tests. A failed GAP complement search is never a certificate of nonsplitting.

The last source expression. Row 28 has order 1536 and expression

$$2^2 \times 4 \cdot 2 : 2 : 2 : 2 : S_3.$$

Left association gives successive orders $16, 32, 64, 128, 256, 1536$. In particular, the direct product at the start is formed before the nonsplit extension. The last four newly generated models complete the required chain from order 64 onwards and its final S_3 complement. The literal reader proves these models by the same predicates as every other row; no special interpretation is introduced to obtain this result.

The proposal search used two elementary improvements. For an abelian requested expression it rejects a subgroup with different cyclic prime-power invariants, after checking abelianness. These invariants are obtained by multiset union for direct products. It also requires the requested action before selecting the first prime-order complement generator. An earlier selection could choose a centralizing involution and overlook a later involution with nontrivial action. Both changes concern proposal efficiency and completeness, and leave the proof predicates unchanged. They do not constitute new structure theorems.

Verification and attribution. The complete literal check passed in 33.710 seconds. It checked all 1369 source joins, the 199 new model proofs, 1747 transports and 1682351 generator edges. The nonsplitting checks include 1376 outside elements and 9472 composite-quotient lift pairs, in addition to the Cayley-graph counts above. Input hashes remained unchanged and stderr was empty. Twenty damaged source or certificate inputs are rejected in both ordinary and optimized Python. The complete optimized replay reproduces the same proof counts. Source

admission is recorded separately by the distinct review, with the extraction qualifications stated above.

The original table, its notation and association convention are due to the source authors. The numerical catalogue and named models are acquired results. The finite transport, extension and lift criteria are reused from the project's accepted common method. The contribution here is their explicit application to every occurrence of this table, with literal certificates and an exact source join. No new conjecture resolution, Lean formalization, historical-byte identity, whole-article completion or compression measurement is claimed. The other large source tables remain in scope.

References

[1] T. Connor and D. Leemans, An atlas of subgroup lattices of finite almost simple groups, *Ars Mathematica Contemporanea* **8** (2015), 259–266. The retained companion table is headed $\mathrm{Sp}(6,2)$ and dated 7 June 2013.